Week 12: Naive Bayes Classifier

ML LAB

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PES1UG23AM309

Purpose: Multinomial Naive Bayes classifier

Result:

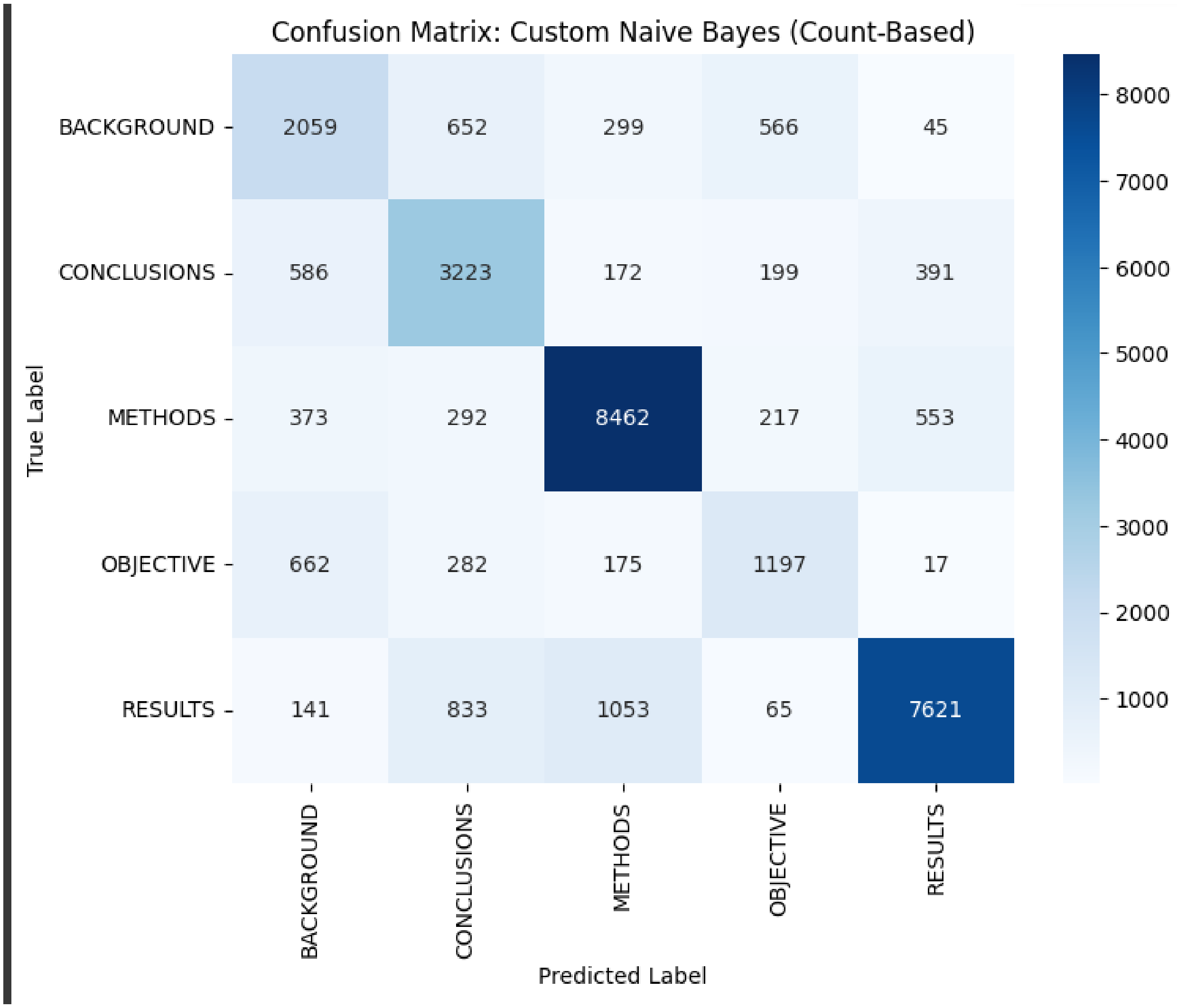
PART A

```
=== Test Set Evaluation (Custom Count-Based Naive Bayes) ===
Accuracy: 0.7487
              precision    recall  f1-score   support

  BACKGROUND       0.54      0.57      0.55      3621
 CONCLUSIONS       0.61      0.71      0.65      4571
     METHODS       0.83      0.86      0.84      9897
   OBJECTIVE       0.53      0.51      0.52      2333
     RESULTS       0.88      0.78      0.83      9713

    accuracy                           0.75     30135
   macro avg       0.68      0.69      0.68     30135
weighted avg       0.76      0.75      0.75     30135

Macro-averaged F1 score: 0.6811
```


Confusion Matrix: Custom Naive Bayes (Count-Based)
True Label
Predicted Label
BACKGROUND
CONCLUSIONS
METHODS
OBJECTIVE
RESULTS
2059
652
299
566
45
586
3223
172
199
391
373
292
8462
217
553
662
282
175
1197
17
141
833
1053
65
7621
8000
7000
6000
5000
4000
3000
2000
1000

PART B:

```
Training initial Naive Bayes pipeline...
Training complete.

=== Test Set Evaluation (Initial Sklearn Model) ===
Accuracy: 0.7257
              precision    recall  f1-score   support

  BACKGROUND       0.64      0.42      0.51      3621
 CONCLUSIONS       0.62      0.61      0.62      4571
     METHODS       0.72      0.90      0.80      9897
   OBJECTIVE       0.74      0.10      0.18      2333
     RESULTS       0.80      0.87      0.83      9713

    accuracy                           0.73     30135
   macro avg       0.70      0.58      0.59     30135
weighted avg       0.72      0.73      0.70     30135

Macro-averaged F1 score: 0.5861

Starting Hyperparameter Tuning on Development Set...
Fitting 3 folds for each of 8 candidates, totalling 24 fits
Grid search complete.

Best Cross-Validation Score (f1_macro): 0.6567
Best Parameters Found: {'nb__alpha': 0.1, 'tfidf__ngram_range': (1, 2)}
```

PART C:

```
Please enter your full SRN (e.g., PES1UG22CS345): PES1UG23AM309
Using dynamic sample size: 10309
Actual sampled training set size used: 10309

Training all base models...
  Training NaiveBayes...
  Training LogisticRegression...
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_c
  warnings.warn(
  Training RandomForest...
  Training DecisionTree...
  Training KNN...
All base models trained.
  Log-Likelihood for NaiveBayes: -986.96
  Log-Likelihood for LogisticRegression: -912.47
  Log-Likelihood for RandomForest: -998.06
  Log-Likelihood for DecisionTree: -1293.33
  Log-Likelihood for KNN: -1489.22
Calculated Posterior Weights: [4.49093784e-033 1.00000000e+000 6.73591246e-038 3.95666039e-166
 3.33456583e-251]

Refitting all models on full sampled training data...
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_c
  warnings.warn(

Fitting the VotingClassifier (BOC approximation)...
Fitting complete.

Predicting on test set...
```

```
Predicting on test set...

=== Final Evaluation: Bayes Optimal Classifier (Soft Voting) ===
Accuracy: 0.7088
              precision    recall  f1-score   support

  BACKGROUND       0.55      0.37      0.44      3621
 CONCLUSIONS       0.61      0.56      0.58      4571
     METHODS       0.71      0.89      0.79      9897
   OBJECTIVE       0.66      0.35      0.45      2333
     RESULTS       0.80      0.81      0.80      9713

    accuracy                           0.71     30135
   macro avg       0.66      0.60      0.61     30135
weighted avg       0.70      0.71      0.69     30135
```

```
Macro-averaged F1 score: 0.6145
```